

Department Overview

- **Department Name:** Department of Biotechnology
- **Faculty:** Faculty of Life Sciences & Informatics
- **Core Academic Pillars:** Built upon five foundational pillars: the students, the faculty, the curriculum, the learning environment, and the civil society served.
- **Core Scientific Focus:** High-demand, rapidly evolving field combining advanced biological sciences with technological application. The curriculum integrates high-precision research, structural field knowledge, and soft skills.
- **Career Horizons:** Prepares graduates for diverse sectors including medicine, clinical laboratory networks, bioethics, education, physical sciences, and life sciences.

1. Bachelor of Science in Biotechnology (BS Biotechnology)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Medical or Pre-Engineering) from any recognized board (or an equivalent qualification) with at least **60% marks**.

Degree Requirements

- **Total Credit Hours:** 135 – 137
- **Total Courses Required:** 42 – 44
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote Total Credits (Theory - Lab).

- **1st Semester:** English-I Functional English (3-0), Pakistan Studies (2-0), Mathematics-I Pre-Calculus (3-0), Organic Chemistry (2-1), Ecology, Biodiversity & Evolution-I (3-0), Cell Biology (2-1). **Total Credit Hours: 17**
- **2nd Semester:** English-II Communication Skills (3-0), Islamic Studies/Ethics (2-0), Biomathematics (3-0), Inorganic Chemistry (2-1), Ecology, Biodiversity & Evolution-II (2-1), Microbiology (2-1). **Total Credit Hours: 17**
- **3rd Semester:** English-III Tech Writing & Presentation Skills (3-0), Introduction to Computer (2-1), Physical Chemistry (3-0), Genetics (2-1), Biochemistry-I (2-1), Introduction to Sociology (3-0). **Total Credit Hours: 18**
- **4th Semester:** Biological Physics (3-0), Probability and Biostatistics (3-0), Analytical Chemistry and Instrumentation (2-1), Biochemistry-II (2-1), Molecular Biology (3-0), Marketing (3-0). **Total Credit Hours: 18**
- **5th Semester:** Introduction to Biotechnology (3-0), Immunology (2-1), Methods in Molecular Biology (1-2), Principles of Biochemical Engineering (2-1), Introduction to Bioinformatics (2-1). **Total Credit Hours: 15**
- **6th Semester:** Genetic Resources and Conservation (3-0), Microbial Biotechnology (2-1), Plant Biotechnology (2-1), Dairy and Food Biotechnology (2-1), Research Methodology (3-0), Elective-I (2-1). **Total Credit Hours: 18**

- **7th Semester:** Genomics and Proteomics (2-1), Health Biotechnology (2-1), Environmental Biotechnology (2-1), Seminar-I (1-0), Elective-II (3-0), Research Project Allocation (0-6). **Total Credit Hours: 13**
- **8th Semester:** Industrial Biotechnology (2-1), Biosafety and Bioethics (1-1), Research Project/Internship (0-6), Elective-III (3-0), Elective-IV (3-0), Seminar-II (1-0). **Total Credit Hours: 18**

Available Undergraduate Electives Pool

Animal Biotechnology, Marine Biotechnology, Radiobiology, Hospital Waste Management, Water and Waste-water Treatment, Nano Biotechnology, Fungal Biotechnology, Pharmaceutical Biotechnology, Biosensors, Biofuels and Bio refineries, Molecular Diagnostics, Cell and Tissue Culture, Virology, and Fermentation Biotechnology.

2. Master of Science in Biotechnology (MS Biotechnology)

Admission Requirements

- **Academic Background:** A BS degree in Biotechnology (or equivalent benchmark qualification) with a CGPA $\geq$ 2.0, OR a minimum of 16 years of education in a relevant life sciences discipline passed with at least **50% marks**.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses Listed:** 24 (*Note: The raw university layout contains a text typo erringly displaying total course count as 24 instead of the actual credit hour pool layout*).
- **Minimum Academic Standing:** CGPA $\geq$ 2.5

Program Schema

- **1st Semester (12 Credits):** Advances in Molecular Biology (3+0), Research Methods in Biotechnology (3+0), Elective-I (3+0), Elective-II (3+0).
- **2nd Semester (12 Credits):** Developments in rDNA Technology (3+0), Biostatistics & Laboratory Mathematics (3+0), Elective-III (3+0), Elective-IV (3+0).
- **3rd Semester (6 Credits):** THESIS-601 Research & Thesis framework commencement.
- **4th Semester (6 Credits):** THESIS-601 Research & Thesis culmination and final defense.

Specialized Research Elective Streams Pool

Graduate students choose coursework from one of four primary research clusters:

- **Stream-I: Human Genetics**
Modern Biotechnology: Principles & Applications, Bioethics/Biosecurity/Biosafety, Advances in Cell and Molecular Biology, Recent trends in Molecular Diagnostics, Advances in Molecular Genetics, Advances in Protein Chemistry, Advances in Animal

Biotechnology, Advances in Biosensor Technologies, Forensic Sciences, Advances in Animal Cell Culture, Advanced Biochemistry, Cellular Signaling, Advances in Health Biotechnology, Advances in Vaccine Research, Molecular Systematics, Pathology, and Animal Physiology.

- **Stream-II: Plant Biotechnology**

Advances in Environmental Biotechnology, Biotechnology Law & Regulation, Molecular Basis of Plant Development, Metabolic Pathways in Plants, Regulation of Gene Expression, Molecular Evolution, Molecular Basis of Plant Breeding, Fungal Biotechnology, Advances in Plant Biotechnology, Advances in Agriculture Biotechnology, Applications of Nano biotechnology, Advances in Plant Tissue Culture, Medicinal Plant Biotechnology, Biopharming in Plants, Advances in Proteomics, Advances in Genomics, Microbial Enzyme Technology, Biological Safety and Risk Management, Plant Pathology, and Plant Microbe Interaction.

- **Stream-III: Bioinformatics**

Advances in Bioinformatics.

- **Stream-IV: Industrial Biotechnology**

Bioprocess Technology, Advances in Pharmacogenomics, Biophysics, Advances in Fermentation Technology, Advances in Immunology, Advances in Microbiology, Advances in Microbial Genetics, Recent trends in Biochemical Engineering, Protein Engineering and Enzyme Technology, Bioremediation and Biodegradation, Biotechnology of Nonrenewable Resources, and Metabolic Engineering and Biofuels.

3. Doctor of Philosophy in Biotechnology (PhD Biotechnology)

Admission Requirements

- **Academic Background:** An HEC-recognized MS degree in Biotechnology or an equivalent postgraduate qualification in a life sciences discipline with a minimum CGPA ≥ 3.0 or at least **70% marks**.

Degree Requirements

- **Total Coursework Credit Hours:** 18
- **Total Courses Required:** 5 – 8 advanced doctoral courses.
- **Minimum Academic Standing:** CGPA ≥ 3.0
- **Doctoral Dissertation:** Full-time, mandatory research work spanning at least two years (36 research thesis credits).

Program Schema

- **1st Semester (9 Credits):** Plant and Agriculture Biotechnology (3-0) (*Core*), Optional Subject-I (3-0), Optional Subject-II (3-0).
- **2nd Semester (9 Credits):** Functional Genomics (3-0) (*Core*), Optional Subject-III (3-0), Optional Subject-IV (3-0).
- **3rd to 6th Semester (36 Credits):** DSS-990 Research & Thesis dissertation.

Available PhD Optional Subjects Pool

Research Methodology, Biostatistics, Agriculture Bioinformatics, Epidemiology, Biofuel and Biorefinery, Radiobiology, Biomaterial, Current Trends in Bioinformatics, Diagnostics, Fungal Biotechnology, Industrial Biotechnology, Marine Biotechnology, Pharmaceutical Biotechnology, Biomedical Informatics, Waste Management, Edible Oil Seed & Health Nutrition, Gene Microarray, Water and Waste Water Treatment, Nanobiotechnology, Bioremediation, Microarray Technology, Microbial Genetics, Plant Biotechnology, Nutrigenomics, Contemporary Plant Biotechnology, Contemporary Animal Biotechnology, Contemporary Agriculture Biotechnology, Protein Microarray, Biology of Oil Seeds & Nutrition, Contemporary Health Biotechnology, Advance Research Reading-III, Research Specialization-I, II, & III, Transcriptomics, Transgenicity & Applications, Bioelectronics & Biosensor, Molecular Epidemiology of Hepatitis, Gene Expression and Replication of Hepatitis Virus, Genetics Diversity of Hepatitis Virus, Stem Cell Biology, Genetic Disorders, Cancer Biology, Antisense RNA technology, Epigenetics, Computational Biology, Pathology, Animal Physiology, Plant Physiology, Animal Pathology, Plant Pathology, Plant Microbe Interaction, Drug Development, and Drug Metabolism.